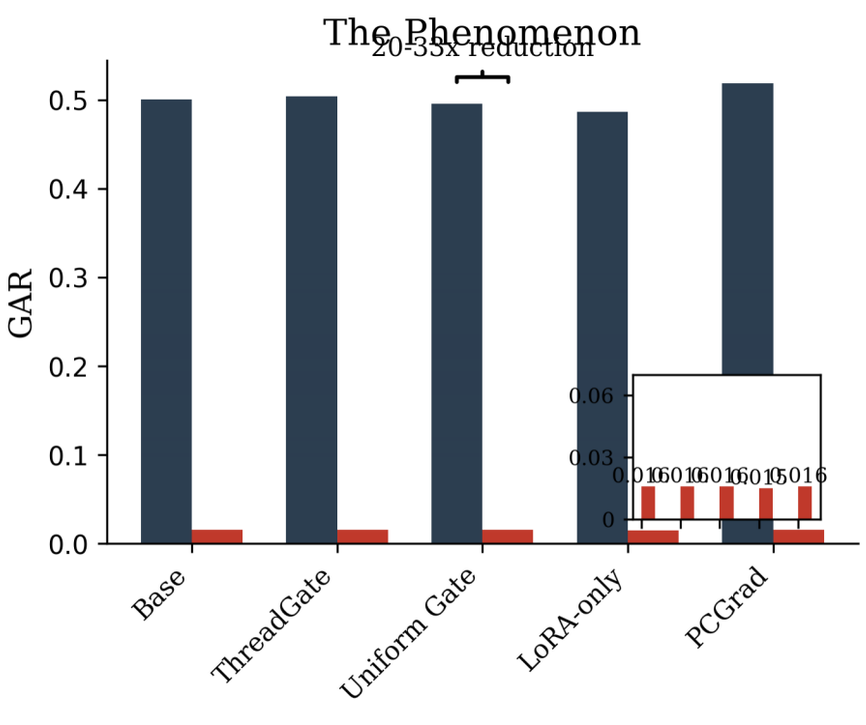
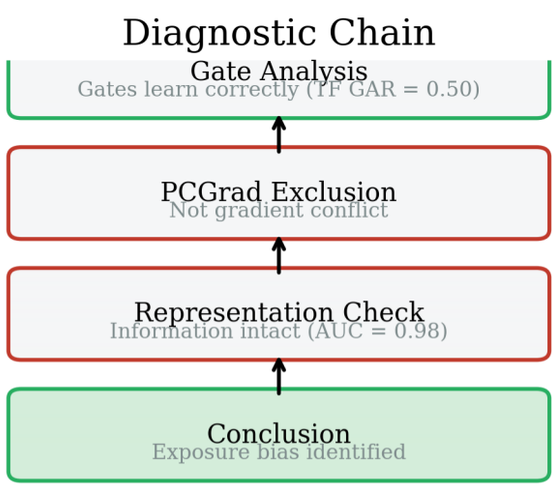
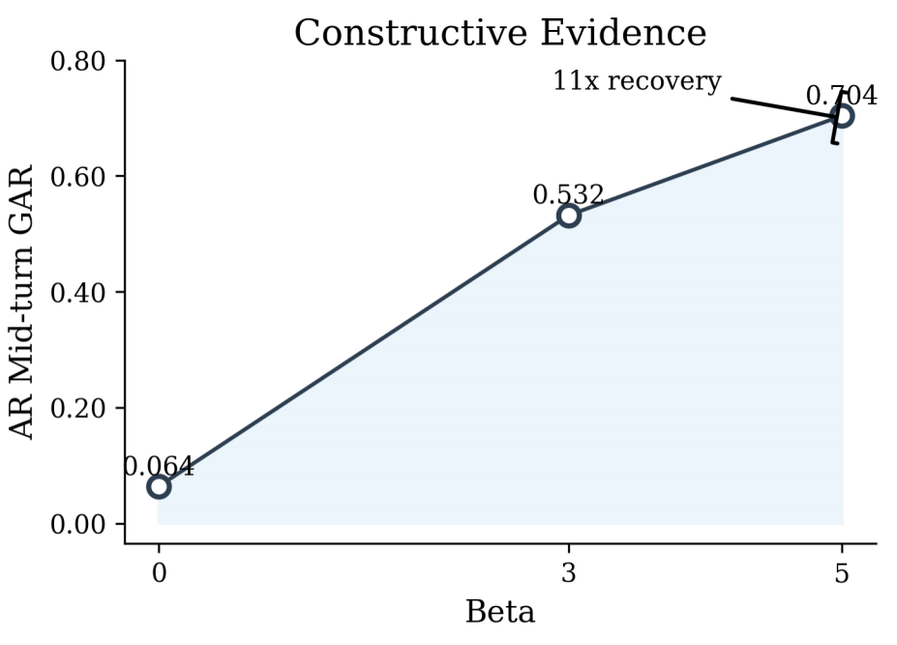

The Phenomenon
20-33x reduction
GAR
0.0
0.1
0.2
0.3
0.4
0.5
0.06
0.03
0
Base
ThreadGate
Uniform Gate
LoRA-only
PCGrad


Diagnostic Chain
Gate Analysis
Gates learn correctly (TF GAR = 0.50)
PCGrad Exclusion
Not gradient conflict
Representation Check
Information intact (AUC = 0.98)
Conclusion
Exposure bias identified


Constructive Evidence
AR Mid-turn GAR
0.00
0.20
0.40
0.60
0.80
11x recovery
0.704
0.532
0.064
0
3
5
Beta